

● EASY

● ADVANCED MATH

Advanced Math

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

ADVANCED MATH

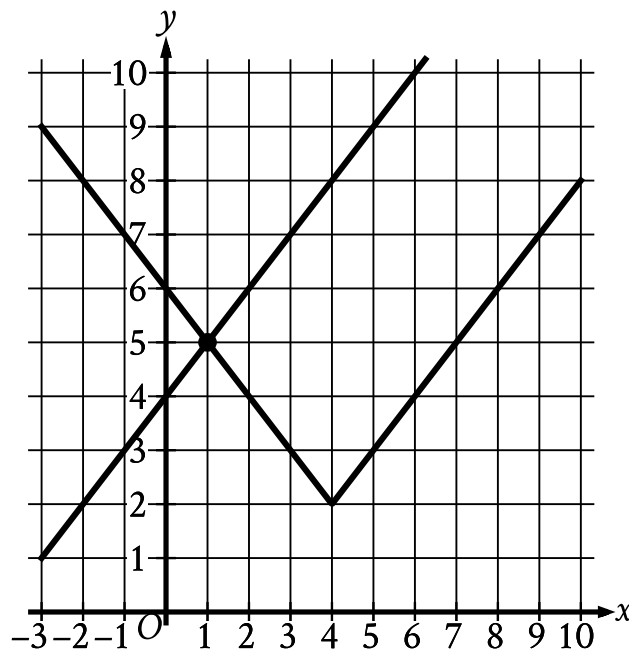
There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p , eliminated at the end of round r , where $r \leq 4$?

A. $p = 15\frac{1}{2}^r$

B. $p = 152^r$

C. $p = 240\frac{1}{2}^r$

D. $p = 2402^r$



- For the absolute value function:
 - Moving from left to right:
 - The function slants sharply down to the point (4 comma 2).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (1 comma 5)
 - (4 comma 2)
 - (6 comma 4)
- For the linear function:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (0 comma 4)
 - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1,5

B. 0,4

C. 1,5

D. 4,2

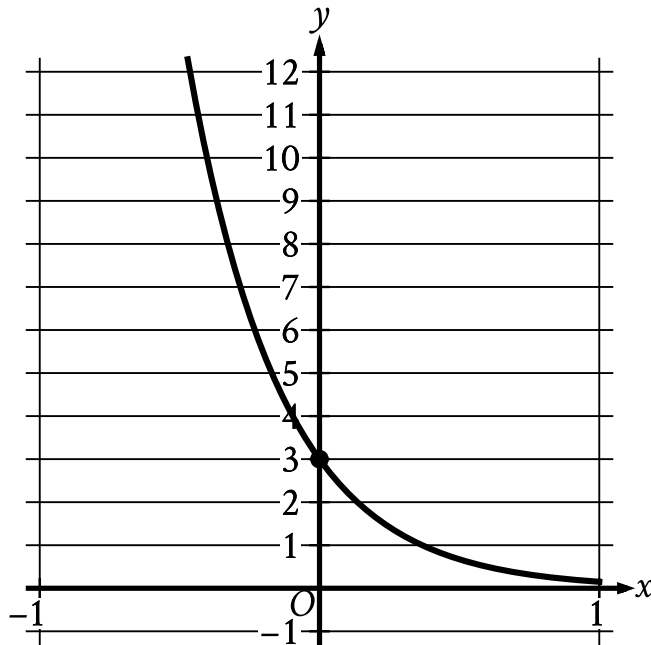
Which expression is equivalent to $11x^3 - 5x^3$?

A. $16x^3$

B. $6x^3$

C. $6x^6$

D. $16x^6$



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends down sharply to the point (0 comma 3).
 - In quadrant 1, the curve trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (0 comma 3)
 - approximately (0.5 comma 0.7)

The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is 0, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$6r = 7s + t$$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

A. $s = 42r - t$

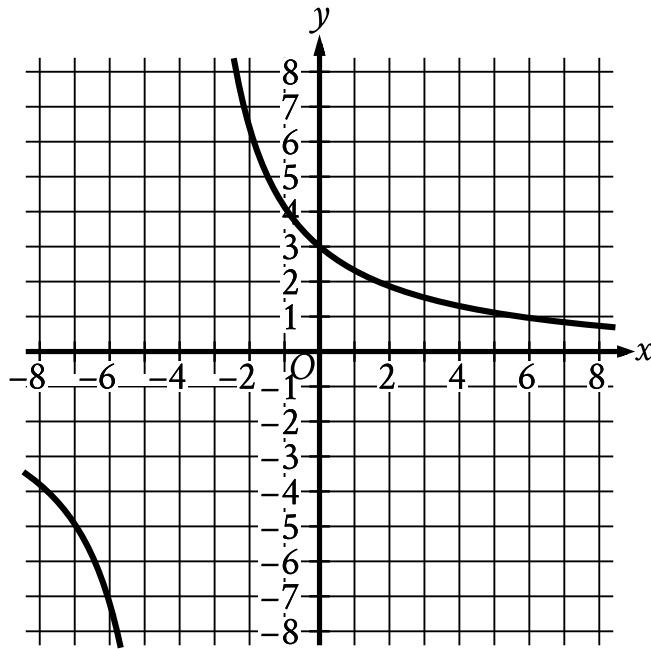
B. $s = 76r - t$

C. $s = \frac{6}{7}r - t$

D. $s = \frac{6r - t}{7}$

Which expression is equivalent to $17x^2 - 100y^2$?

- A. $17x - 2yx - 50y$
- B. $17x - 2yx + 50y$
- C. $17x - 10yx - 10y$
- D. $17x - 10yx + 10y$



- For the first curve:
 - Moving from left to right:
 - The curve is in quadrant 3.
 - As x decreases, the curve approaches the line y equals 0.
 - As x increases, the curve approaches the line x equals negative 4.
- For the second curve:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - As x decreases, the curve approaches the line x equals negative 4.
 - As x increases, the curve approaches the line y equals 0.
- The 2 curves pass through the following points:
 - approximately (negative 7 comma negative 4.9)
 - (0 comma 3)
 - approximately (7 comma 0.8)

The graph of $y = f(x)$ is shown in the xy -plane. The value of $f(0)$ is an integer. What is the value of $f(0)$?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = 76$$

$$y = x^2 - 5$$

The graphs of the given equations in the xy -plane intersect at the point (x, y) . What is a possible value of x ?

A. $-\frac{76}{5}$

B. -9

C. 5

D. 76

Which expression is equivalent to $13x^2 - 7x^2$?

A. $-91x^2$

B. $6x^2$

C. $20x^2$

D. $40x^2$

At the time of posting a video, a social media channel had 53 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the video was posted?

A. $n = 53^d$

B. $n = 53(2)^d$

C. $n = 53\frac{1}{2}^d$

D. $n = 53^2 + d$